Problem statement

Built a **production-oriented Streamlit app** that takes **labeled** and **unlabeled** CSVs, **auto-detects domains/URLs** from messy text, and **classifies each domain as *legitimate* or *malicious***.
It combines:

- **A machine-learning ensemble (XGBoost + Random Forest over TF-IDF n-grams + structural features), and**
- **A hardened security layer (whitelist of genuine bank domains, brand/TLD policy rules, typosquatting & homograph detection, suspicious-subdomain patterns, and risk scoring).**

The output is a high-signal table with predictions, confidence, and **explainable risk factors** (e.g., wrong TLD for brand, suspicious subdomain like login/verify, character substitutions like sb1 vs sbi, etc.) plus summary charts.

Dataset used & logic followed (with diagram)

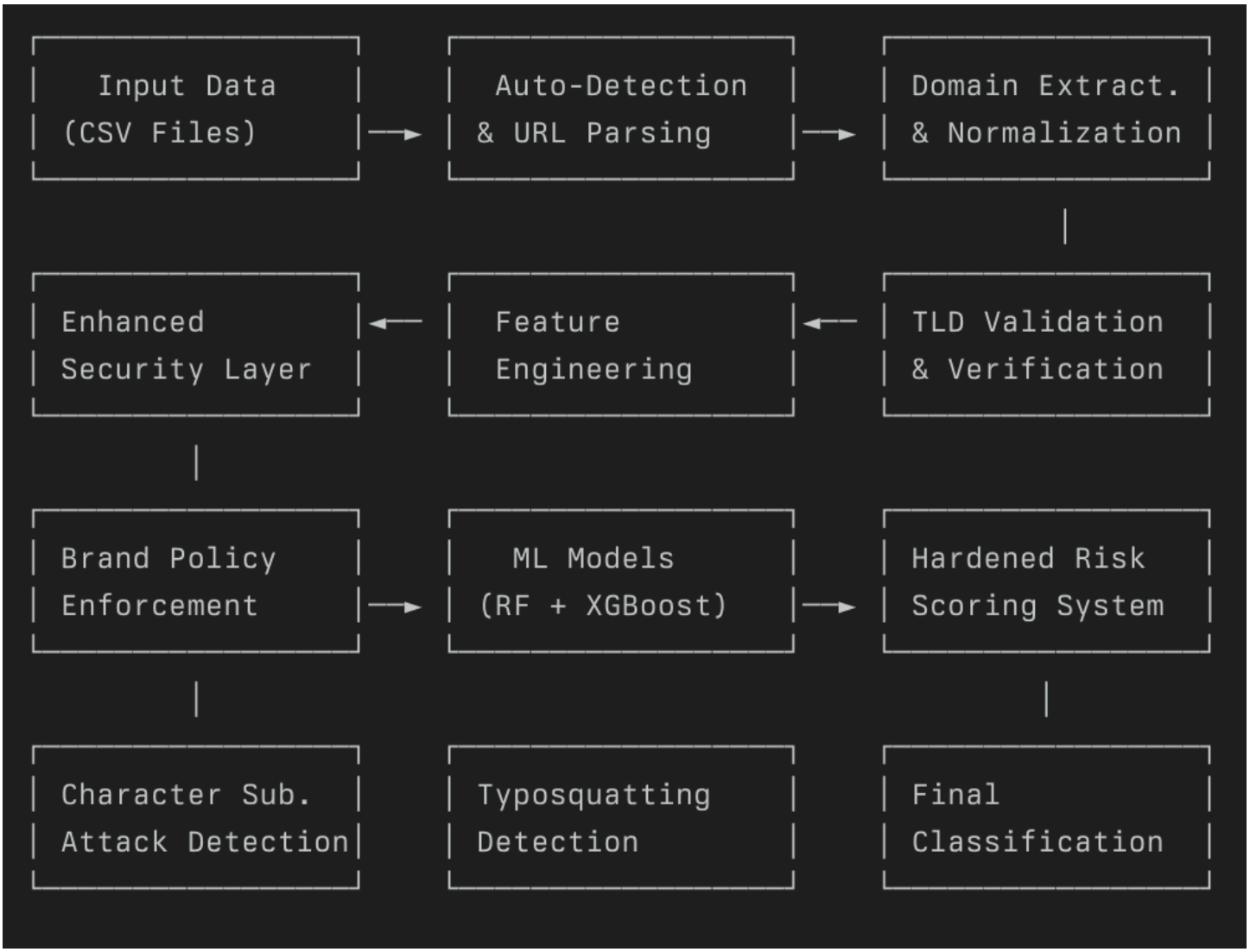
Input datasets

Two CSVs (uploaded in sidebar):

1. **Labeled training data** – must contain at least one column with domains/URLs and one label column (names auto-detected). Labels are normalized to {malicious, legitimate} via mapping (e.g., phishing/spam/malware/1 → malicious, benign/0/safe → legitimate).
2. **Unlabeled prediction data** – at least one column with domains/URLs (auto-detected).

Built-in reference data (hardcoded):

- LEGITIMATE_DOMAINS_CSV: a large whitelist of **verified bank domains** (India + a few international).
- BRAND_TLD_POLICIES: per-brand **official TLDs** and **forbidden TLDs** (e.g., sbi → official .sbi, forbid .in/.org/.xyz/...; icicibank/hdfcbank/axisbank/kotak → official .com, forbid .in/.xyz/.top/...).
- HIGH_RISK_TLDS: risky ccTLDs/gTLDs (e.g., tk, ml, cf, ga, xyz...).
- Homograph maps, digit-letter substitutions, and suspicious subdomain tokens (login, secure, verify, update, account, ...).



Key implementation details

Auto-detection & extraction

- **Column detection** (smart_column_detection, auto_detect_and_extract_domains): scores columns by domain-likeness; extracts domains from **any** text column (emails, URLs, free text) with regex + cleaning.
- **Label detection** (detect_label_column_enhanced): names like label/class/category/type/status weigh higher; content heuristics confirm binary labels.

Feature engineering

- **Structural (~25 features)** (extract_enhanced_structural_features_vectorized_v2):
  - Lengths/digits/letters/specials for SLD & full domain
  - Subdomain depth, 3rd/4th-level flags, overall domain levels
  - **TLD intelligence**: is_genuine_tld, tld_type (govt/banking/other)
  - Ratios (digit/share, vowel ratio), char diversity, **entropy**
  - Counts of suspicious tokens; homograph characters
- **Text features** (create_ngram_features_memory_efficient_enhanced):
  - **Char n-grams** (2–4) and **word n-grams** (1–2) using TF-IDF
  - Memory-aware: chunking, adjustable max_features (sidebar), optional sampling for huge data

Model training & inference

- **Ensemble** (ProductionEnsembleClassifier):
  - **XGBoost** (hist tree method, depth/child/colsampling tuned; binary vs multi auto-handled)
  - **Random Forest** (balanced class_weight, tuned depth/leafs)
  - Average predicted probabilities across models
  - Batch/partial-fit pathway for very large matrices
- **Baseline metrics block** (kept in code): simple RF over char 3–5 n-grams with a train/test split to quickly compute **Accuracy** and **F1** (see §3).

Hardened security layer (post-ML)

- **Whitelist** (check_legitimate_domain_enhanced) — **highest priority**: whitelisted domains are directly classified as **legitimate** with high confidence.
- **Brand/TLD policies** (check_brand_policy_violation) — flags official brands using wrong TLDs or risky TLDs → strong risk increment (often **override** ML).
- **Typosquatting** (detect_character_substitution_attacks, normalize_domain_aggressive) — insertion/deletion/substitution (e.g., sb1, hdfc-bank, homoglyphs).
- **Suspicious subdomains** (SUSPICIOUS_SUBDOMAINS) — e.g., login.verify.update.
- **Risk score** (calculate_hardened_legitimacy_score) — aggregates factors; thresholds (70=High, 90=Critical). Tracks whether **ML was overridden**.

Streamlit UX (sidebar options)

- Include 3-level domains, **Chunk size**, **Max n-gram features**, **Batch size**, **Parallel jobs**, **Sampling** & **Sample size**
- **Confidence threshold** (marks “used_for_training” in semi-supervised loops)
- Toggles: **Use XGBoost**, **Use Random Forest**, **Enable Hardened Security Laye**

Performance Metrics

Model Evaluation Results:

Accuracy: **90.28%** - Overall prediction correctness

F1 Score: **94.81%** - Excellent balance of precision/recall for malicious class

Performance Optimizations:

Memory Efficiency: Sparse matrix operations, chunked processing

Scalability: Handles 1M+ records with configurable batch sizes

Processing Speed: Parallel processing (1-4 cores), optimized feature extraction

Risk Classification Thresholds:

0-50: Legitimate/Low Risk

50-70: Moderate Risk

70-85: Suspicious

85-90: High Risk

90-100: Critical Threats

Security Enhancement Metrics:

Brand Policy Violations: Automatic detection of wrong TLD usage

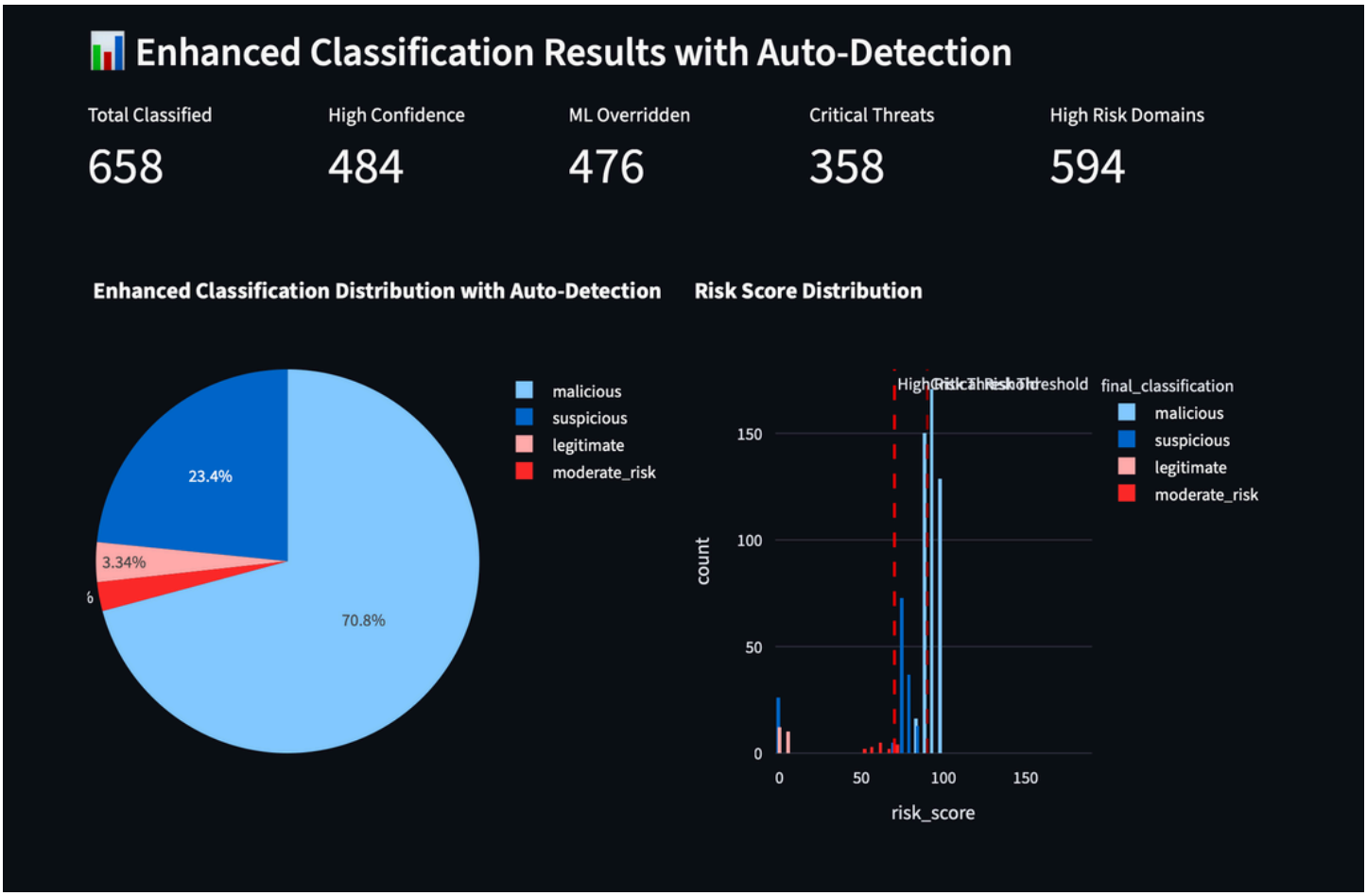
Character Substitution: Multi-algorithm similarity scoring

Whitelist Coverage: 1000+ verified legitimate domains

ML Override Rate: Security layer can override ML predictions

Sample Output

	domain	sld	full_domain	tld_type	is_genuine_tld	predicted_label	final_classification	confidence	final_confidence	risk_score
0	canarabannk.com	canarabannk	canarabannk	genuine	☑	legitimate	malicious	0.7934	0.88	88
1	indianbank-mypay.live	indianbank-mypay	indianbank-mypay	standard	☐	malicious	malicious	0.9616	0.95	96.156
2	bandhaanbank.online	bandhaanbank	bandhaanbank	standard	☐	legitimate	suspicious	0.5477	0.6	0
3	idffcfirstbank.net	idffcfirstbank	idffcfirstbank	genuine	☑	malicious	suspicious	0.6058	0.75	75
4	statebankofindia.live	statebankofindia	statebankofindia	standard	☐	legitimate	suspicious	0.5777	0.6	0
5	axisbank.com	axisbank	axisbank	banking/corporate	☑	legitimate	legitimate	0.9272	0.95	5
6	yesank.link	yesank	yesank	standard	☐	legitimate	suspicious	0.824	0.75	75
7	canaraabank.co	canaraabank	canaraabank	standard	☐	legitimate	malicious	0.8756	0.88	88
8	yesbank.in	yesbank	yesbank	genuine	☑	legitimate	legitimate	0.8595	0.95	5
9	customer.idbibank.info	idbibank	customer.idbibank	standard	☐	malicious	suspicious	0.7956	0.7956	79.5565
10	kotak-support.link	kotak-support	kotak-support	standard	☐	malicious	malicious	0.8854	0.8854	88.5426
11	kotak.info	kotak	kotak	standard	☐	legitimate	malicious	0.8598	0.95	95
12	state6ankofindia.link	state6ankofindia	state6ankofindia	standard	☐	malicious	suspicious	0.8134	0.8134	81.3415
13	otp.rblbank.in	rblbank	otp.rblbank	genuine	☑	malicious	moderate_risk	0.6803	0.6803	68.0319
14	indianbank-account.online	indianbank-account	indianbank-account	standard	☐	malicious	malicious	0.9532	0.95	95.3187
15	ax1sbank.co	ax1sbank	ax1sbank	standard	☐	malicious	malicious	0.754	0.9	90
16	unionbankfoindia.org	unionbankfoindia	unionbankfoindia	genuine	☑	malicious	moderate_risk	0.541	0.541	54.1007



Additional Relevant Details

Enhanced Security Features:

1. Brand Policy Enforcement Database:

2. Character Substitution Detection:

Homoglyph Detection: Cyrillic/Greek characters that look like Latin

Digit Substitution: 0→o, 1→i, 3→e, 5→s detection

Keyboard Adjacency: Common typos (q→w, a→s)

Levenshtein Distance: Multi-algorithm similarity scoring

3. High-Risk TLD Monitoring:

Blacklisted TLDs: .xyz, .top, .click, .tk, .ml, .cf, .ga

Often abused for phishing campaigns

4. Suspicious Subdomain Patterns:

Keywords: login, secure, verify, update, account, auth

Especially dangerous on wrong TLDs

Technical Architecture:

Memory Management:

Sparse matrix operations for 1M+ feature vectors

Chunked processing (configurable 5K-100K chunks)

Garbage collection optimization

Batch model training for large datasets

Auto-Detection Capabilities:

Full URLs: https://banking.axisbank.com/login

Domain Names: axisbank.com, suspicious-site.xyz

Email Domains: user@icicibank.com → extracts icicibank.com

Mixed Text: Extracts all domains from any text format

IP Addresses: 192.168.1.1:8080

Ensemble Learning Architecture:

Random Forest: 50 trees, balanced class weights

XGBoost: 100 estimators, histogram tree method

Prediction Averaging: Weighted ensemble combination

Confidence Thresholding: Configurable (default 85%)